

The Membrane Voltage

Neurons are excitable cells

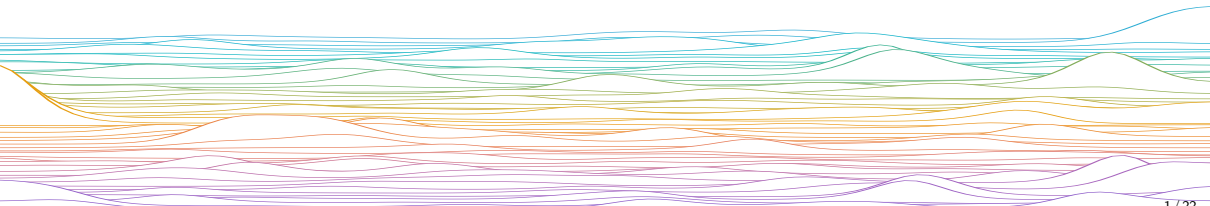
Computational Neuroscience

University of Bristol

M Rule

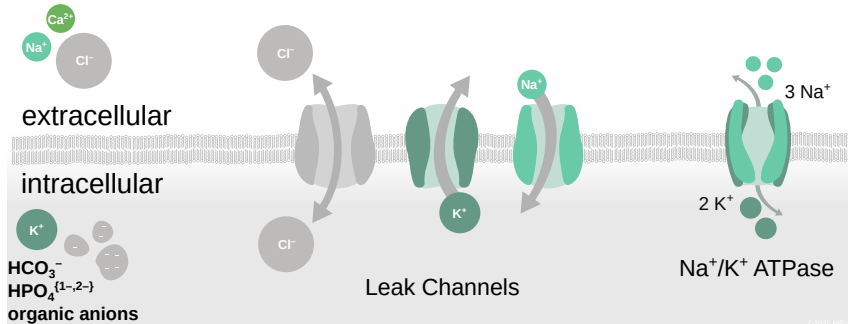
Learning outcomes:

- ▶ Leak channels, $\text{Na}^+/\text{K}^+\text{ATPase}$
- ▶ Nernst potential
- ▶ Effective circuit for passive membrane
- ▶ Detail the role of Na^+ , K^+ , Cl^- ions in the membrane voltage.



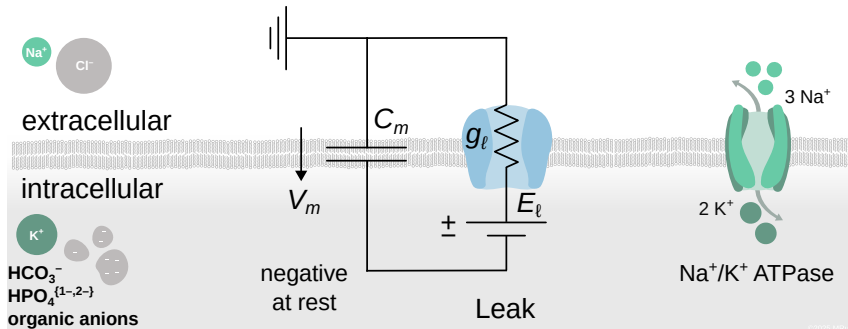
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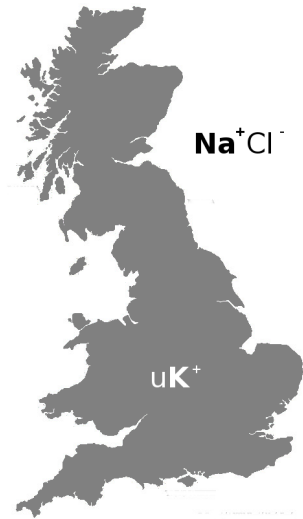
- ▶ They maintain a voltage difference across their cell membrane
- ▶ Charged ions with differing intracellular/extracellular concentrations → chemical “batteries”
 - Effective voltage depends on ion's charge & concentration difference
 - Powers voltage dynamics for computation & communication
 - Active pumps set and maintain concentration differences



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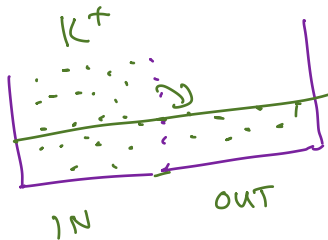
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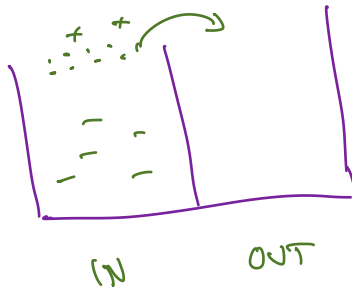


draw a picture of the balance of energy (voltage) and entropy (diffusion) fluxes

Diffusion/entropy



Voltage/energy



Consider: ion C with charge z , concentrations $[C]_{\text{out}}$, $[C]_{\text{in}}$.

Moving ion inside:

- ▶ **ΔEnergy** zFV
- ▶ **ΔEntropy** $R \ln \left(\frac{[C]_{\text{out}}}{[C]_{\text{in}}} \right)$

Electrical work

Chemical work

Balance?

$$\Delta \text{Energy} = T \cdot \Delta \text{Entropy}$$

$$zFV \propto RT \ln \left(\frac{[C]_{\text{out}}}{[C]_{\text{in}}} \right)$$

$\vdots$

$$V = \frac{T}{z} \frac{R}{F} \ln \left(\frac{[C]_{\text{out}}}{[C]_{\text{in}}} \right)$$

$$= \frac{1}{z} E_0 \ln \left(\frac{[C]_{\text{out}}}{[C]_{\text{in}}} \right)$$

$$\frac{R}{F} = \frac{\text{Gas constant}}{\text{Faraday constant}}$$

Nernst potential

Often use $T = 310.15 \text{ K}$ (37°C), such that $E_0 := TR/F \approx 26.7 \text{ mV}$

- ▶ T : Temperature (Kelvin, assume $\sim 310.15^\circ\text{K}$, i.e. $37^\circ\text{C} + 273.15$)
- ▶ R : Ideal gas constant $\approx 8.3145 \text{ J K}^{-1} \text{ mol}^{-1}$
- ▶ F : Faraday's constant $\approx 96,485 \text{ C mol}^{-1}$

Example: Chloride reversal potential in mature pyramidal neuron

$z = -1$: Ion charge

$E_0 \approx 26.7$

$[\text{Cl}^-]_{\text{out}}$: 110 mM

$[\text{Cl}^-]_{\text{in}}$: 7 mM

E_{Cl^-} : Cl^- reversal potential (voltage)

$$E_{\text{Cl}^-} = \frac{1}{z} E_0 \ln \left(\frac{[\text{Cl}^-]_{\text{out}}}{[\text{Cl}^-]_{\text{in}}} \right) = (-26.7 \text{ mV}) \ln \left(\frac{110 \text{ mM}}{7 \text{ mM}} \right) \approx 73.5 \text{ mV}$$

Goldman–Hodgkin–Katz (GHK) voltage equation

In practice, multiple leak channels set resting voltage, and channels may be permeable to multiple ions; To calculate reversal potentials with multiple ionic fluxes you need to use the *Goldman–Hodgkin–Katz (GHK) voltage equation*.

$$E_m = E_0 \ln \left(\frac{P_{\text{Na}}[\text{Na}^+]_{\text{out}} + P_{\text{K}}[\text{K}^+]_{\text{out}} + P_{\text{Cl}}[\text{Cl}^-]_{\text{in}}}{P_{\text{Na}}[\text{Na}^+]_{\text{in}} + P_{\text{K}}[\text{K}^+]_{\text{in}} + P_{\text{Cl}}[\text{Cl}^-]_{\text{out}}} \right)$$

$$e^{E_m/E_0} = (\text{permeability} \cdot \text{concentration}) \text{ weighted average} \dots = \frac{\sum_i w_i e^{z_i(E_i/E_0)}}{\sum_i w_i}$$

$$w_i = P_i \cdot \begin{cases} [i]_{\text{in}} & \text{if } z_i \text{ positive} \\ [i]_{\text{out}} & \text{if } z_i \text{ negative} \end{cases}$$

Some of the voltage and ligand (i.e. neurotransmitter) gated channels we will cover are also permeable to multiple ions (at least in part), and require a similar treatment. But, each ion channel can be reduced to an *effective* reversal potential and conductance.

end

appendix

Jargon

depolarize v_m increases

repolarize v_m decreases towards resting potential

hyperpolarize v_m is driven below resting potential

Sign conventions

Let $v_m = v_{\text{interior}} - v_{\text{exterior}}$ denote a neuron's membrane voltage

A positive current applied to a neuron makes v_m _____?

- ▶ *Engineer*: increase
- ▶ *Electrophysiologist*: decrease!

Compromise:

- ▶ **inward current** makes v_m more positive
- ▶ **outward current** makes the membrane voltage more negative

Nernst: Intuitive

Consider ion C^z and two equal, small volumes in the exterior/interior of the cell with extracellular/intracellular concentrations $[C]_{\text{out}}$ and $[C]_{\text{in}}$, respectively. Let $N = [C]_{\text{out}} + [C]_{\text{in}}$ be the total number of ions, and define $p = [C]_{\text{in}}/N$, the portion $p \in [0, 1]$ inside the cell.

The Nernst potential describes a relationship between p and membrane voltage V that minimizes free energy.

$$F = \text{energy} - T \cdot \text{entropy} \quad (1)$$

We will consider a voltage clamp experiment, which fixes V and allows p to relax to equilibrium. To minimize free energy in p , differentiate F and set to zero $\frac{d}{dp}F(p) = 0$, yielding the relation:

$$\frac{d}{dp}\text{energy} = T \frac{d}{dp}\text{entropy}. \quad (2)$$

Nernst: Intuitive

Entropy: Diffusion wants to equalize interior and exterior concentrations. The entropy is how many questions we need to ask to know whether a given ion is inside or outside the cell. We use entropy of a Bernoulli distribution for a coin-toss with $\text{Pr}(\text{success} \sim \text{inside}) = p$:

$$\frac{d}{dp} \text{entropy} \propto \frac{d}{dp} \{-p \ln(p) - (1-p) \ln(1-p)\} = \ln\left(\frac{1-p}{p}\right). \quad (3)$$

Energy: The energy used to move p ions with charge z into a cell with electric potential difference V is

$$\frac{d}{dp} \text{energy} \propto \frac{d}{dp} \{zVp\} = zV. \quad (4)$$

Equating (3) and (4) per (2) gives the Nernst equation:

$$V = \frac{T}{z} \alpha \ln\left(\frac{1-p}{p}\right), \quad (5)$$

This V is the reversal potential for a given ion concentration $[C]_{\text{out}}/[C]_{\text{in}} = (1-p)/p$.

To recover physical units, use $\alpha = \frac{\text{Ideal gas constant}}{\text{Faraday's constant}} = \frac{R}{F}$.

end

